



Quantifying the degree of coral bleaching using digital photographic technique

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ABSTRACT

A simple, non-destructive and objective method is developed and evaluated to quantify bleaching or colour change in coral in terms of its percent whiteness. Digital photos of coral colony were taken with a photoquadrat marked with black and white bands around a metal frame to serve as colour references. Photos taken were then converted to a black and white grey-scale and intensities of the coral colour were determined with reference to this grey-scale using computer image analytical technique. A completely bleached coral gave an estimated percent whiteness that is close to 100%, an estimate that agrees well with the visual image of a bleached coral. The percent whiteness estimates were found not to be significantly affected by different levels of irradiance and turbidities under simulated conditions. Colour change in seven common Hong Kong coral species were further evaluated in situ and shown to exhibit an increasing trend of percent whiteness from January (winter) to April 2012 (spring), reflecting an increasing degree of bleaching. Using *Goniopora lobata* as an experimental species, estimated percent whiteness was shown to be negatively correlated with its zooxanthellae density and chlorophyll a concentration. This suggests that this method could allow repeated monitoring of individual coral colony over time to detect subtle variations in its colour as a general assessment of its health. Furthermore, reference percent whiteness of different coral species can be standardized for comparison of colour change over time to identify the critical level of percent whiteness that is associated with bleaching. Thus, this method should prove to be an easy and objective tool for rapid field assessment of coral health.

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1. Introduction

Coral bleaching event, marked by whitening of corals due to the loss of their symbiotic zooxanthellae or the photosynthetic pigments of these symbionts (Brown, 1997), has increasingly been reported around the world in recent years (Glynn, 1983; Hoegh-Guldberg, 1999; Wilkinson, 2000; Oliver and Palumbi, 2011). Numerous stressors have been shown to induce coral bleaching, including elevated and decreased seawater temperature, solar (UV) radiation, reduced salinity and also bacterial and other infections (Glynn, 1993; Brown, 1997; Dove and Hoegh-Guldberg, 2006; see Berkelmans, 2009 for review). Large-scale mass bleaching event was observed in Panama, Florida and other Caribbean sites in 1982–1983 (Glynn, 1984). In 1998, a series of mass bleaching events occurred all over the world, destroying 16% of the world's coral reefs (Wilkinson, 1998). Most recently, another mass bleaching event occurred in 2010 with very significant part of Southeast Asian reefs being seriously affected (e.g. Doshi et al., 2012; Yucharen and Yeemin, 2012).

Monitoring of coral bleaching events has thus become a major focus of many reef studies. Methods have been developed to monitor these

events worldwide. However, many of these methods rely on subjective visual assessment of coral colour change. Descriptive terms like “seriously bleached” or “slightly bleached” have been used. This qualitative approach, however, would not be amenable to a more detailed assessment and comparative study of the severity of the bleaching events as differences in coral colour change between species, sites, seasons and years estimated by different researchers cannot be quantified (Fitt et al., 2001). More recently, a popular method has been introduced with the use of Coral Health Chart (Siebeck et al., 2006) with six different colour intensities on four types of colours (green, brown, red and reddish brown). This method is easy to use and allows divers to observe the colour of corals by placing this chart next to the corals to observe coral colour intensity. However, since there are only six colour intensities for each colour, resolution of colour difference is limited. In some cases, differences in coral colour cannot be readily noticeable if the coral colour falls between two intensity levels. Furthermore, this coral health chart method is still subjective, depending highly on the sensitivity of the diver/observer to colours. Moreover, some corals have distinctive colours (e.g. yellow or blue) other than the four colours provided in the chart so that change in colour intensity becomes difficult to assess. Other more objective and quantitative methods have also been introduced (Edmunds et al., 2003; Winters et al., 2009) but these require the use of relatively expensive apparatus (e.g. an underwater single-

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lens reflex camera with dual strobe or external LED light source) with principal component analysis which would not be easy for an amateur diver to manage and use to report bleaching events in his locality. Some of the designs for quantitative assessment were also applicable only in a laboratory setting (Koren et al., 2008). Therefore, a more practical, easy and nonetheless objective and precise method to quantify coral colour change or bleaching in the field or laboratory is very much needed. With the rapid development of digital photography, quantifying colour change using digital photography is becoming more popular and affordable. Techniques using digital photography have now been widely applied to evaluate colour changes in vegetation cover of forest ecosystem as a way to assess phenological events (Sonnentag et al., 2012; Keenan et al., 2014).

Along this same principle of digital photography, we developed a method to allow quantitative analysis of the degree of bleaching or colour change in corals. In this method, a photoquadrat is mounted with a digital camera housed in a diving case and the images taken are analysed using computer image analytical techniques. This method makes use of black and white markers on the quadrat as references. Photos taken with the quadrat are converted to a black and white grey-scale and intensities of the coral colour are then determined with reference to this grey-scale using computer image analysis.

This technique was tested on the coral *Goniopora lobata* as well as on other coral species in Hong Kong which experience seasonal change in colour. Samples of *G. lobata* were also collected bimonthly and examined in the laboratory for zooxanthellae density count and chlorophyll a (Chl a) concentration determination. Changes in the colour intensity of the corals were correlated with changes in their zooxanthellae density and Chl a concentration.

This technique is shown to be objective, and is independent of coral colours and colour sensitivity of the divers. It can easily be used in the field as well as in the laboratory. This technique is also a non-destructive method that can be used repeatedly on the same coral colony over time for a quick assessment of its colour change.

2. Methods

2.1. Design of photoquadrat

A 18 × 13 cm stainless steel frame (from Nikonos™ Close Up Unit) was mounted with a digital camera in an underwater diving case (Sony CyberShot WX-1, Japan). Black and white PVC insulating tapes (2 cm width) were cut and wrapped around the frame in alternate order (Fig. 1A). Altogether, there were 15 black tapes and 15 white tapes around the frame that served as black and white references for image intensity calibration.

2.2. Photo image analysis

Photos taken were analysed with the computer image analysis program Image-pro Plus (Version 6.0, Media Cybernetics, USA). As a first step, photos were converted to 8-bit grey scale. A colour intensity calibration was then carried out using those black and white reference points around the photoquadrat. The actual intensities of these reference points were first determined. These intensity readings were then reset to a range of 0 to 255 (8-bit), with the lowest intensity reading representing black colour being set as 0, and the highest intensity reading representing white colour being set as 255. This calibrated intensity range was used to assess the colour intensity of the coral taken in the same photo. Using 100 random points generated by the image analysis program (Fig. 1B), the intensity of each random point over part of the coral was recorded within the same range as the reference points from 0 to 255. Bleaching intensity was then expressed as a percentage of whiteness using the

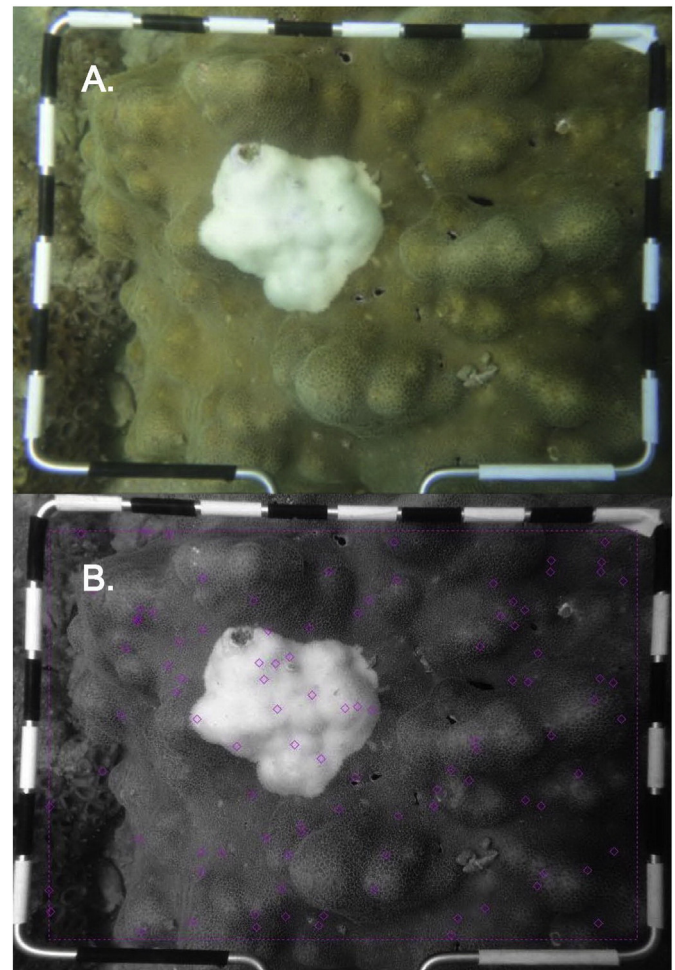


Fig. 1. A. Photo taken in situ of *Porites* sp. with a 18 × 13 cm stainless steel frame (from Nikon Close-up Kit) marked with black and white PVC insulating tapes. B. Same photo converted into a grey scale using the computer image analysis software. Random points are first marked on the black and white PVC tapes to generate the reference intensity scale, with 0 representing extreme black, and 255 representing extreme white colour. Additional 100 random points (shown in purple rhombus) are generated over the surface of the coral colony, with the intensity of each point measured with respect to the grey scale of 0 to 255. [See text for more detailed description of the calculation for percent whiteness of the coral colony.](#) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

following equation:

Percent whiteness (bleaching intensity) = Average intensity of all random points over coral sample / 255 × 100%.

In the extreme case, a completely bleached coral, with colour intensity as white as the white reference point, should have a 100% or close to 100% whiteness or bleaching intensity. On the other hand, a completely black coral, as dark as the black reference point, should have a percent whiteness of 0. This latter value is unlikely to be obtained as no coral will be as black as the black reference point. Photos taken in the trial run to normalize the technique (see below), as well as in the field to test the application of this technique (see below) were analysed using the approach described above.

2.3. Trial run and normalization of the technique

Trial runs for the application of this technique were carried out on different coral species with *Porites* sp. as a typical example since it is easy to find different shades of colour change within a single colony of

this species. In an extreme case, part of the colony could be completely bleached while the other part appears normal, i.e. showing a generally brownish colouration. Furthermore, this technique could potentially be affected by levels of irradiance and light quality (light colour spectra), reflecting different light conditions at different water depths or at different times of a day. It could also be affected by water turbidity which in turn could affect both the quantity and quality of light that penetrates through the water column. Sensitivity of this technique to these different conditions, in terms of its ability to estimate percent whiteness of a coral, was therefore evaluated.

2.3.1. Colour intensity of normal and bleached sections of the coral *Porites* sp.

Photo of a *Porites* sp. colony (Fig. 1A) that was partly bleached and partly normal was taken in Wu Pai, northeastern Hong Kong. One hundred random points were generated on the whole photoquadrat (Fig. 1B), as well as separately on the normal and bleached sections of the colony to allow comparison of the bleaching intensity (percent whiteness) of these different sections.

2.3.2. Colour intensity of the coral *G. lobata* under different exposure settings and contrasts

A *G. lobata* colony collected from the same study site was placed in a self-circulating water table under ambient condition and was mounted on a modified PVC plate with alternate black and white PVC insulating tapes (2 cm width) around its perimeter. This background plate was to replace the photoquadrat (i.e. photo frame) described above. This modification was necessary to apply this technique in a small tank. Photos of this coral colony were taken under different exposure settings with auto-metering (from -1 EV to $+1$ EV, in 7 steps) (Fig. 2A) to simulate underwater conditions with different levels of irradiance. A photo taken under normal exposure was then edited using photo editing software (ACDSee Pro 2, ACD Systems Inc., USA) to adjust the photo

contrast that ranged from -75 to $+75$ in 7 steps (Fig. 2B) to simulate underwater conditions with different turbidity levels.

2.3.3. Colour intensity of the coral *Porites* sp. under different colour sources

A *Porites* sp. colony was placed in a small cubic tank ($12 \times 12 \times 12$ cm) under indoor condition and was mounted on a modified PVC plate similar to the one described above. Photos of this coral colony were taken under an external light source (Holga VL-36 LED Video Light, China) with different semi-transparent colour plates placed in front of the light source acting as colour filters. Photos with ambient, red, yellow, green and blue lights were compared to determine the percent whiteness of the coral under different light conditions. These trial runs evaluated whether this photographic technique is sensitive to colour change, simulating for example underwater conditions with red light being absorbed faster than blue light along the water depth.

2.3.4. Colour intensity of the coral *Platygyra acuta* under different levels of irradiance

P. acuta colonies were placed in a self-circulating water table under ambient condition and mounted on modified PVC plates similar to those described above. Photos of these coral colonies were taken at different times of a day in three different sunny days to reflect different levels of irradiance. The level of irradiance was recorded and percent whiteness of the coral in the photo taken was estimated in each case. These trial runs determined whether this technique is sensitive to change in the levels of irradiance at different times of a day.

2.4. Field application of the technique on other coral species

To evaluate the applicability of this technique to quantify colour change in other coral species, especially those with different growth forms, photos of several coral species were taken in Wu Pai in January and April 2012 to compare winter and spring colour change among these corals. At least three individuals of the same coral species were

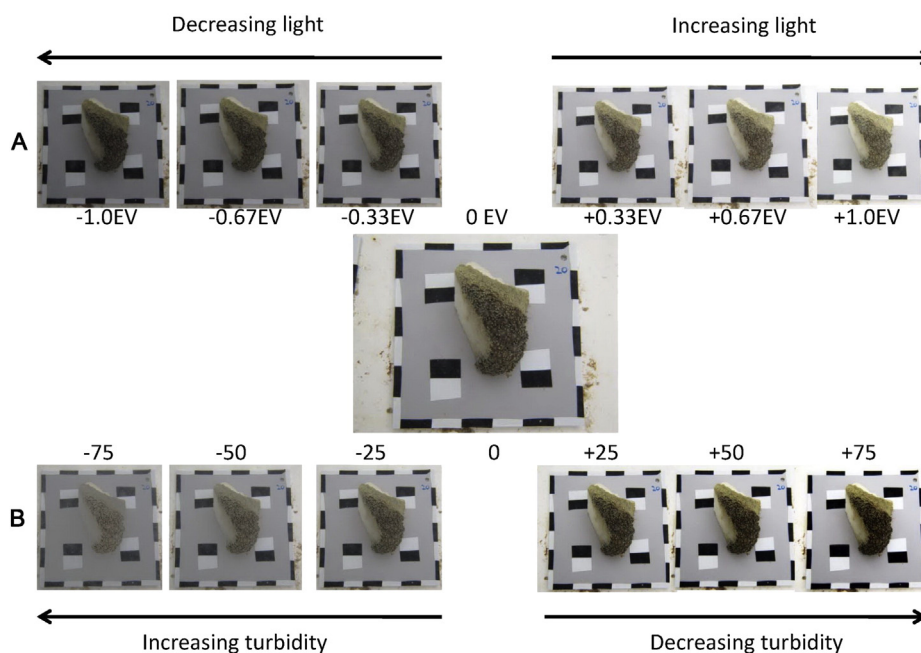


Fig. 2. A. Photos of the same piece of coral colony (*Goniopora lobata*) under different exposure compensations (-1.0 EV to $+1.0$ EV) set in the camera to simulate different levels of irradiance. 0.0 EV represents the normal auto setting. B. Photo taken under normal auto setting was set to represent 0 level of contrast, with decreasing (lower) and increasing (higher) contrasts (-75 to $+75$) adjusted using the computer image analysis software. Increasing level of contrast simulates decreasing turbidity (i.e. clearer water) and decreasing level simulates increasing turbidity.

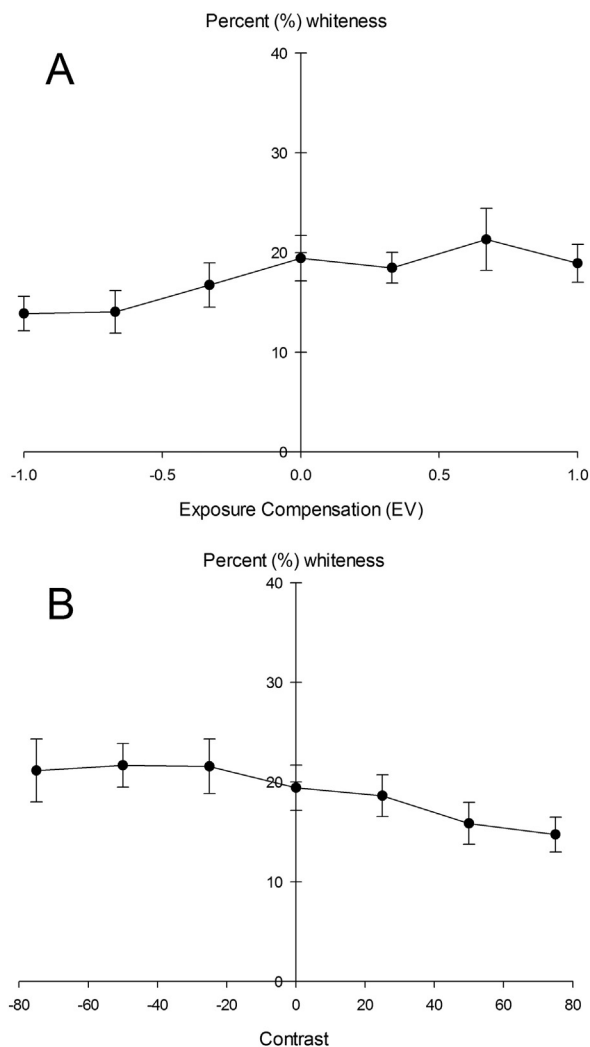


Fig. 3. A. Mean ($\pm$ SE) percent whiteness of *Goniopora lobata* calculated from photos taken under different exposure compensations (-1 EV to $+1$ EV in 7 steps) shown in Fig. 2A. Changes in mean percent whiteness are not statistically significant (One-way ANOVA, $F = 1.35$, $p = 0.24$). B. Mean ($\pm$ SE) percent whiteness of *G. lobata* calculated from photos adjusted in different contrasts (-75 to $+75$ in 7 steps), as shown in Fig. 2B. Changes in mean percent whiteness are not statistically significant (One-way ANOVA, $F = 0.90$, $p = 0.50$).

chosen haphazardly in the field every time and a total of seven coral species were examined.

2.5. Evaluation of the use of percent whiteness to estimate health of the coral *G. lobata*

The coral *G. lobata* was used as a test species to evaluate how percent whiteness can be used to assess coral health in terms of its zooxanthellae density and Chl *a* concentration.

2.5.1. Collection and photo taking of samples

Twenty *G. lobata* colonies (10 visually assessed to be normal and 10 to be bleached or discoloured) were examined every month from April 2011 to April 2012 from Wu Pai, northeastern Hong Kong. Visually normal and visually paler colonies were collected to compare their zooxanthellae density and Chl *a* concentration. Underwater photos of each colony were first taken with a marked photoquadrat as described above. Zooxanthellae density and Chl *a* concentrations of the corals

were taken in the laboratory in order to assess their correlations with their calculated percent whiteness (bleaching intensity).

2.5.2. Tissue extraction, sample preservation and determination of zooxanthellae count and Chl *a* concentrations

In every other month during the sampling period, a small fragment of each of the coral samples was collected. These samples were brought back to the laboratory submerged in sea water. Tissue was then extracted from these samples for zooxanthellae and Chl *a* analysis using Water-Pik extraction method (Waterpik®, USA) (Johannes and Wiebe, 1970). One polyp from each sample was extracted in 50 ml filtered seawater. A fluorescent lamp (NightSea FL-1 FLASH Light™, USA) was used to ascertain that all coral tissue within the corallite has been removed. The 50 ml coral tissue solution was then homogenized and centrifuged at 12000 rpm for 30 min (Himac CR21G centrifuge, Hitachi, Japan). The pellet obtained was washed in 10 ml filtered seawater and further homogenized for 30 s.

Three sub-samples (about 0.4 mm³ each) from each sample extraction were mounted on a Neubauer haemocytometer to count the zooxanthellae density under the light microscope. All spherical zooxanthellae in four grids of the haemocytometer were counted and the density of zooxanthellae was expressed in terms of per unit area of the coral surface ($\times 10^6$ cells cm⁻²).

Chl *a* pigments were extracted from 2 ml of the 10 ml homogenized coral tissue mixed with 8 ml of 90% acetone in darkness. After 24 h in the dark, each mixed solution was centrifuged at 12000 rpm for 15 min (Hitachi Himac CR21G centrifuge, Japan). The supernatant was then transferred into a glass cuvette and its absorbance taken at 750, 664, 647 and 630 nm in a UV-Vis spectrophotometer (Hitachi U-2001, Japan), with 90% acetone used as a blank. Chl *a* concentration was expressed using the equation of Jeffrey and Humphrey (1975).

2.6. Data analysis

All statistical analyses were performed using SPSS 15.0 for Windows (SPSS Inc., USA) with significant level (p value) set at 0.05. The data were first tested for normality using Kolmogorov-Smirnov test and homogeneity of variance using Levene's Test of Equality of error variances (homogeneity test for ANOVA/repeated measure ANOVA).

To evaluate the effect of exposure compensation or contrasts (simulating different light and turbidity conditions respectively) and the effect of colour sources (simulating different underwater conditions), the mean percent whiteness for each photo setting was determined. Differences in this parameter were compared using One-way ANOVA with exposure compensation, contrasts or colour as the fixed factor. Post hoc tests (Tukey's HSD or Tamhane's T2 tests when equal variances were violated) were performed to find the similarity groupings among them. The relationships between mean percent whiteness and levels of irradiance were examined using linear regression analysis.

Seasonal changes in the mean percent whiteness of *G. lobata* as well as that of other coral species were assessed using One-way ANOVA or student's *t*-test. The relationships between mean percent whiteness and zooxanthellae density and Chl *a* concentration per cm² coral surface area were examined using linear regression analysis. Seasonal relations of these parameters with mean percent whiteness of *G. lobata* were also assessed. Outliers in the linear regression analysis were identified by robust regression and outlier removal method (Alma, 2011).

3. Results

3.1. Percent whiteness calculation

Taking the example of the *Porites* sp. colony (Fig. 1A) with part being normal (brown) and part being bleached, percent whiteness of the whole colony, bleached and normal sections of the colony were estimated. Percent whiteness of the normal and bleached parts of the

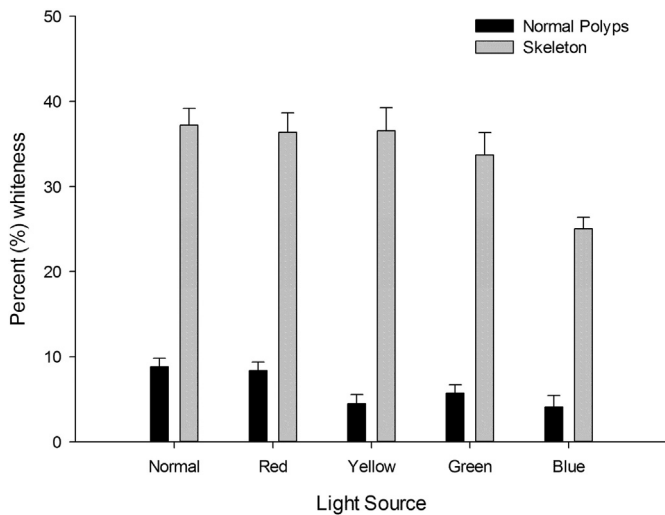


Fig. 4. Mean ($\pm$ SE) percent whiteness for normal polyps and skeleton of *Porites* sp. calculated from photos taken under different light sources. The mean percent whiteness calculated is statistically significantly different (One-way ANOVA, Normal Polyps: $F = 6.66$, $p < 0.01$; Skeleton: $F = 4.64$, $p < 0.01$).

colony was found to be $35.7 \pm 2.2\%$ and $96.3 \pm 1.2\%$ (mean $\pm$ SE) respectively, while percent whiteness of the whole colony was calculated to be $35.1 \pm 1.5\%$. The percent whiteness could thus reflect a general impression of the coral colour. The higher the percent whiteness, the greater is the extent of the white colour and the higher is the bleaching intensity.

3.2. Normalization of the technique

Photos taken underwater are affected by ambient water condition (e.g. turbidity) as well as light (including light levels and angle of incident light). The possible effects of these variables on the assessment of coral colour intensity were evaluated under simulated conditions.

3.2.1. Effect of simulated changes in light condition (exposure compensation) on percent whiteness estimation

Colonies of *G. lobata* from photos taken under different exposure compensation settings [7 photos from -1.0 EV to $+1.0$ EV] showed the percent whiteness to range from 13.9% (-1 EV) to 18.9% ($+1$ EV) (Fig. 3A), with a mean ($\pm$ SE) of $17.6 \pm 1.1\%$. The differences among these percent whiteness estimates are not statistically significant (One-way ANOVA, $F = 1.35$, $p = 0.24$).

3.2.2. Effect of simulated changes in turbidity (contrasts) on percent whiteness estimation

Under different contrast values from -75 to $+75$, the mean ($\pm$ SE) percent whiteness estimated ranged from $21.2 \pm 2.2\%$ at contrast -75 to $14.8 \pm 1.7\%$ at contrast $+75$ (Fig. 3B). The differences among these percent whiteness estimates are not statistically significant (One-way ANOVA, $F = 0.90$, $p = 0.50$).

3.2.3. Effect of simulated changes in colour sources on percent whiteness estimation

A colony of *Porites* sp. from photos taken under an external light source with different semi-transparent colour filters showed the percent whiteness of normal polyps and skeleton to range from $4.10\% \pm 1.34\%$ (mean $\pm$ SE) (blue) to $8.81\% \pm 1.02\%$ (mean $\pm$ SE) (Ambient) and $25.02\% \pm 1.38\%$ (mean $\pm$ SE) (blue) to $37.22\% \pm 1.95\%$ (mean $\pm$ SE) (Ambient), respectively (Fig. 4). The differences among these percent whiteness estimates are statistically significant (One-way ANOVA, Normal Polyps: $F = 6.66$, $p < 0.01$; Skeleton: $F = 4.64$, $p < 0.01$) with blue colour showing significantly lower percent

whiteness estimated for both normal polyps and skeleton (Tukey's HSD, $p < 0.05$).

3.2.4. Effect of levels of irradiance on percent whiteness estimation

Although a significant (ANOVA, $F = 14.72$, $p < 0.01$, $N = 162$) negative linear relationship could be found between the percent whiteness of *P. acuta* colonies and their respective levels of irradiance measured (Fig. S1A), only $<10\%$ of the variability could be explained by this relationship. Furthermore, when data of extreme irradiance ($>1350 \mu\text{mol s}^{-1} \text{m}^{-2}$) were excluded from the data set, no significant linear relationship could be found between the percent whiteness of *P. acuta* colonies and their respective levels of irradiance measured (Fig. S1B) (ANOVA, $F < 0.01$, $p = 0.96$, $N = 114$).

3.3. Colour change in common coral species at different seasons

For all the common coral species examined, a consistent trend could be observed (Fig. 5). All these corals showed a lower percent whiteness in early winter (January 2012) that ranged from 10 to 29%, and an increase in percent whiteness in early spring (April 2012) that ranged from 21 to 45%. Among these corals, *Porites lutea* showed the largest fluctuation in percent whiteness with 24.3% increase, whereas *Favia speciosa* (= *Dipsastraea speciosa*) showed the least (11.1% increase). A trend could also be observed with respect to the coral growth forms, with submassive forms tending to have lower percent whiteness in both seasons (massive to submassive growth forms: 18.9% to 11.0% in January 2012; 38.3% to 23.2% in April 2012).

3.4. Relationship between percent whiteness and health of corals

3.4.1. Percent whiteness vs. zooxanthellae density per cm^2 coral surface area

A significant (ANOVA, $F = 7.10$, $p = 0.02$, $N = 12$) negative linear relationship between percent whiteness and zooxanthellae densities per cm^2 coral surface area of *G. lobata* colonies can also be found (Fig. 6). The seasonal trend, however, was less obvious. Nonetheless, the percent whiteness recorded among the colonies tended to be highest in February 2012 and lowest in October and December 2011.

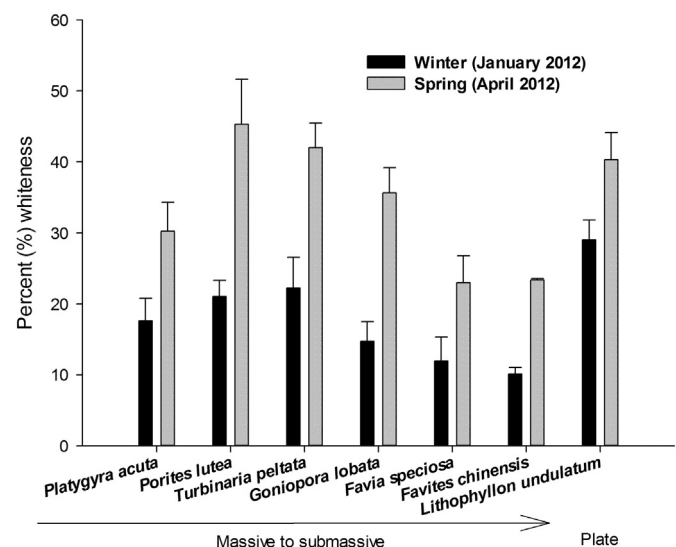


Fig. 5. Significant changes in mean ($\pm$ SE) percent whiteness of seven common Hong Kong scleractinian coral species in January 2012 (winter) and April 2012 (spring) (Two-way ANOVA; Species: $F = 4.86$, $p < 0.01$; Season: $F = 27.76$, $p < 0.01$; Species $\times$ Season: $F = 0.55$, $p = 0.74$).

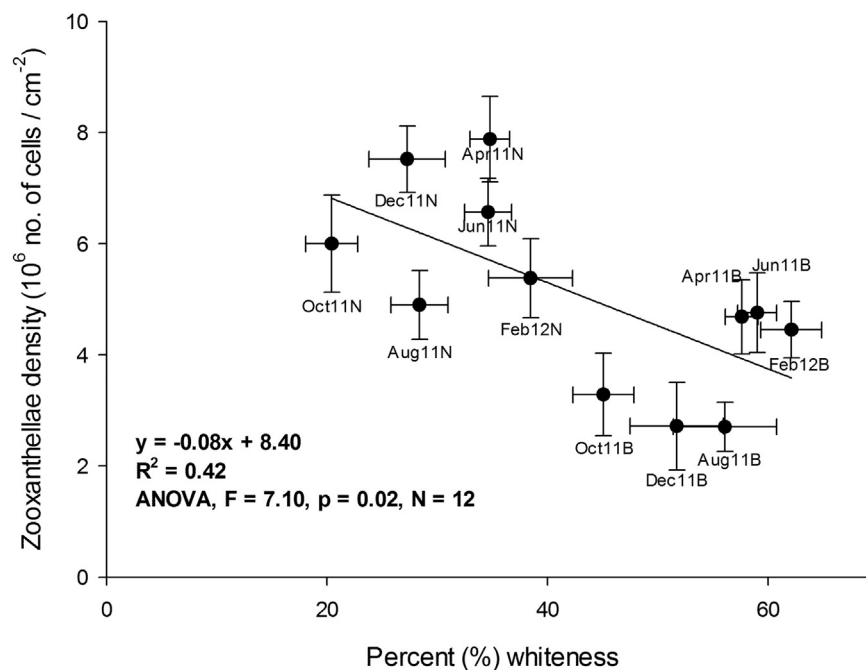


Fig. 6. Bimonthly changes in mean percent whiteness of *Goniopora lobata* from April 2011 to February 2012 and their relationship with averaged zooxanthellae densities per cm² surface area of the coral colonies. Each data point represents mean $\pm$ SE for both zooxanthellae density and percent whiteness of 10 visually normal (N) corals, and 10 visually bleached (B) corals measured in the date indicated.

3.4.2. Percent whiteness vs. Chl *a* concentration per cm² coral surface area

G. lobata colonies that were paler in colour tended to exhibit higher percent whiteness and the linear relationship between percent whiteness and the Chl *a* concentrations per cm² coral surface area was statistically significant (Fig. 7). Coral colonies tended to be paler in colour during summer (August) so visually paler vs. visually normal colonies may not always be clearly distinguishable. The Chl *a* data for visually paler colonies obtained in August (Aug11N) were identified as an outlier and were removed from the regression analysis.

4. Discussion

It is becoming more and more important to be able to assess environmental impacts on coral reef communities objectively (Porter and Ouida, 1992; Hodgson, 1999; Loya, 2004). Coral bleaching is a significant indicator of the health of corals under threat. With the great improvement in digital photography, underwater photo-taking becomes easy and popular and high resolution digital images can easily be obtained even by amateur divers with simple tools. This simple, non-destructive, rapid and objective way to quantify the degree of bleaching or colour change of corals over time that is developed here, utilizing digital imagery, should therefore be very useful if large scale, long term monitoring of impacts on coral reefs is to be carried out.

4.1. Measure of coral colour in terms of percent whiteness

Using the present method, digital photos taken with a series of alternating white and black markers are first converted to a grey scale of black and white that ranges from an intensity of 0 (completely black) to 255 (completely white). The colour of a coral is then defined relative to this scale as the percent whiteness or bleaching intensity. Realistically, 0% whiteness cannot be obtained because corals are never completely black in colour. On the other hand, 100% or close to 100% whiteness is possible, as demonstrated in the example of a bleached *Porites* sp. colony examined in this study. Under extreme bleaching

condition, the colour of the bleached part of *Porites* sp. was estimated to have 96.3% whiteness, suggesting that bleached colour of the coral can reasonably resemble the colour of the white PVC tape used as the reference colour in the photo-image.

In the field, coral bleaching is easy to observe using naked eyes if the extent of bleaching is intense, and the loss of colour is sharp and distinct. However, living corals in any coral community, even of the same species, will exhibit colour in a range of intensities at any one point in time. These subtle variations in colour are simply part of the seasonal change and may not always indicate that these corals are suffering from bleaching. What is deemed to be the normal colour intensity of a coral is thus relative and may need to be evaluated for each species in each locality or region. The application of the present approach to calculate coral percent whiteness takes no a priori assumption of what constitutes a normal colour intensity of the coral. It allows comparison to be made among different coral colonies of the same species, as well as among different species of corals, over spatial and temporal scales. The most common colour intensity can be obtained as a reference and any deviation, and the extent of this deviation, from the "normal" colour can be estimated and compared in terms of percent whiteness.

4.2. Effects of extreme conditions in estimating coral percent whiteness

Ambient water condition (e.g. turbidity) as well as light (including light levels and angle of incident light) can readily affect the quality of an underwater photo. Photos of the same object taken at different times are unlikely to be taken under identical conditions. As photos are used to compare colour change in corals over time, it is important to address the sensitivity of photo quality, and its subsequent use in computer image analysis, to ambient condition under which the photo is taken. It is conceivable to address this question experimentally by taking photos of the same coral colony in an aquarium, whereby the quality of the water could be changed gradually by adding silt to increase its turbidity, or to change the level of irradiance and the angle of incident light to simulate different light conditions under different cloud covers

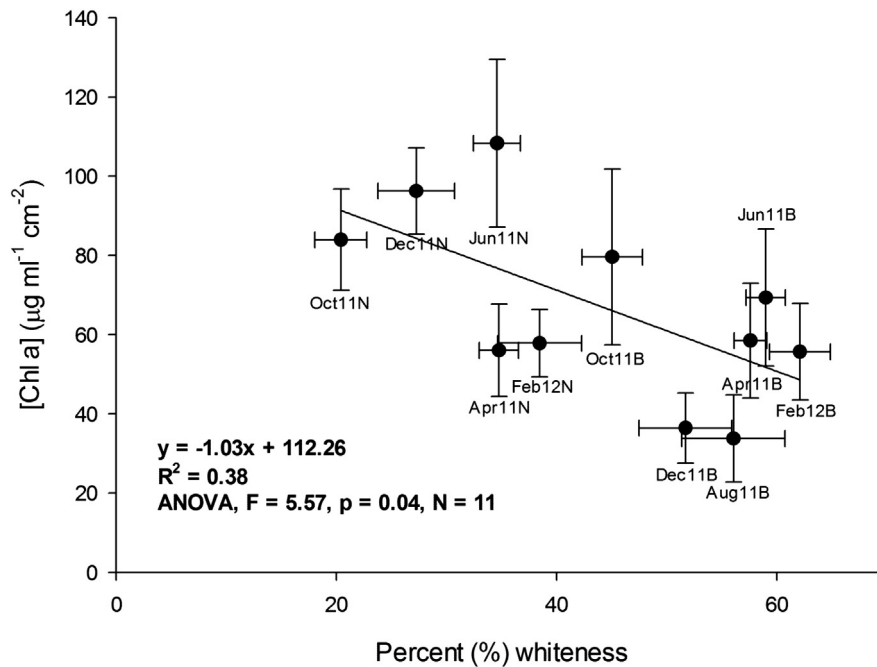


Fig. 7. Bimonthly changes in mean percent whiteness of *Goniopora lobata* from April 2011 to February 2012 and their relationship with averaged Chl *a* concentrations per cm² surface area of the coral colonies. Each data point represents mean $\pm$ SE for both Chl *a* concentrations and percent whiteness of visually normal (N) corals, and visually bleached (B) corals measured in the date indicated. Data for Aug11N were identified as outliers and were removed from the analysis.

or times of the day or season. However, corals are sensitive to turbidity and light conditions. Changes in these conditions could affect their behaviour, e.g. extension of their polyps and/or tentacles, hence would introduce additional variability that could affect the colour of the colony itself. In order to address the question of photo quality, exposure compensation and photo contrast were employed instead to simulate variable light and turbidity conditions respectively in order to avoid adding uncertainty that is due to behavioural changes in the corals. Exposure compensation can readily be manipulated in a digital camera, and photo contrast, in an image editing software.

Variable simulated light and turbidity conditions did not appear to affect percent whiteness estimate for corals. However, it is understandable that under extremely turbid situation, it may be more difficult to obtain an accurate estimate of the percent whiteness of the corals, relative to that obtained under normal condition. Under the extreme conditions, resolution and details of the photos will be lost and hence the original colour of the coral will not be accurately depicted in the photo. There is no ready solution to this. Nonetheless, this is a problem only if comparisons are being made on photos taken on the same coral that is exposed to very contrasting turbidity conditions seasonally. The problem is not serious under less extreme conditions. In Hong Kong, the water is usually turbid as the sedimentation rate is high and could vary between 10 to 200 mg cm⁻² day⁻¹ after storm (Wong, 2001). The extremely turbid conditions, however, may not last for very long (<1 week) after a heavy storm. Colour intensities of several Hong Kong corals were successfully evaluated over time in the present study, indicating that under relatively high turbidity, this photographic method is robust and readily applicable to quantify coral colour change.

4.3. Changes in intensity of coral colours at different seasons

As an application of the present technique, seasonal changes in the colour intensity of seven Hong Kong coral species were assessed. An increasing trend of percent whiteness in early spring (April 2012) was obtained with an average increase of 14.6% in coral percent whiteness from January 2012. This was correlated with a change in temperature in the study site (Wu Pai), with temperature dropping to ~15 °C in February and March 2012 compared to 18 °C in January 2012. Corals thus

lost some of their colour intensity in response to this drop in temperature. Although the temperature has increased back to 20 °C in April, there was a time lag in coral response thus the percent whiteness remained high in April. Corals need time to recover from the cold temperature to gain back energy and nutrients for zooxanthellae synthesis (Furla et al., 2011). Although the overall trends are similar, the percent whiteness of “normal” colour in January 2012 for different coral species is different. Reference percent whiteness for each coral species has to be standardized for future comparison of coral colour change to identify the level of percent whiteness that can be considered as normal or bleached for these corals.

4.4. Comparison with other approaches to quantify coral colour change

Coral species have different natural colourations, e.g. brown, green or blue. Maguire et al. (2003) and Edmunds et al. (2003) have applied similar colour intensity scoring in a 0–255 scale to convert the colour spectral values of RGB (red, green and blue) of a coral image into a single measure using Principal Component Analysis (PCA). This approach can certainly give a more precise description of the colour change in a coral. However, conversion of different colours into a grey scale, as employed in the present approach, would have also considered differences in the colour intensities exhibited by different colours, e.g. brown colour would give a darker shade (lower percent whiteness) compared with a green colour. Any subsequent change, e.g. from brown to green, or from dark brown to light brown, could also be reflected accordingly in the percent whiteness estimate. This is evident from the data obtained on different coral species evaluated in this study. Thus, the present approach is simpler to use, without having to resort to the application of more elaborate multivariate statistics.

In a global scale, remote sensing techniques have been used to assess the extent of coral bleaching (Myers et al., 1999; Joyce and Phinn, 2003; Stambler and Dubinsky, 2005; Hochberg et al., 2006). These techniques also employed photographic approach to measure spectral reflectance. The whole colour channel profile was initially used. This was subsequently simplified by spectral analyses at 675 nm to correlate with the coral pigment content. While the utility of remote sensing approach is no doubt very valuable (Myers et al., 1999; Andréfouët et al., 2003,

2004), the tools needed to carry out such study are, however, expensive. Take for example, the price of a hand held in situ spectral reflectance system is very high (~US\$5000; Hochberg et al., 2006), and will not be affordable to most local coral monitoring teams especially in Southeast Asia where corals are most abundant. The present approach made use of a simple metal frame (photoquadrat) mounted on a digital camera with diving case. Image analysis programmes are also readily available in the internet. It is therefore relatively inexpensive and should not be difficult to use as a popular tool for both scientific and amateur divers in monitoring their own local coral reefs.

Coral Health Chart has become a common technique (Coral Watch, University of Queensland, 2002) to assess the degree of coral bleaching all over the world. It has been adopted in Reef Check, a popular reef monitoring programme involving many volunteers to conduct rapid reef surveys worldwide (Reef Check, USA). The present approach has the advantage over the Coral Health Chart in being able to generate continuous data (i.e. percent whiteness) as compared with the categorical data from the chart. This being the case, it should be able to detect even very subtle change in coral colour in situ. Besides, it is free from the influence of colour sensitivity of the observer. Quantitative, continuous data generated are more amenable to more rigorous statistical analysis and can thus further increase the objectivity in evaluating the intensity of coral colour change or the degree of bleaching. It takes around 2 min to estimate the percent whiteness of a coral from a photo taken using this technique. This approach can therefore be easily applied in rapid assessment of the general health of corals in a coral reef.

4.5. Estimation of coral percent whiteness as a tool to assess coral health

Estimate on coral percent whiteness can even be more meaningful if it could be used to provide a rough assessment of the coral health. These estimates were found to correlate more significantly with seasonal change in zooxanthellae density than with Chl a concentration. It is comparable to other studies where coral zooxanthellae density was found to have a greater correlation with season than Chl a content (Siebeck et al., 2006). This observation may give some insight on the mechanisms that allow corals to survive under extreme temperatures. Corals may expel their symbiotic zooxanthellae under high or low temperatures (Brown, 1997) and hence, a greater drop of zooxanthellae density under extreme temperatures. Some studies, on the other hand, found a significant negative relationship between colour intensity within the red channel and Chl a concentration (Winters et al., 2009). This is thought to be site and species specific as some zooxanthellae may have a more sensitive thermal susceptibility than the host corals and hence their pigments may have been lost before them (Lesser, 2011). As mechanisms involved in coral bleaching differ among corals and presumably also among localities, users of the technique presented in the present study should be encouraged to explore more to find the specific relationship of the percent whiteness of their corals and their zooxanthellae density or Chl a concentrations. If more corals could be assessed in the same way, then a general picture of the health of the coral communities can also be obtained.

4.6. Potential limitations of the technique

The present technique relies on two references (black and white) to calibrate the colour intensity of corals. It is easy to use with digital cameras, and the cost for making the photoquadrat is also minimal. It can readily be used in the field (or even in the laboratory albeit with some modifications) and therefore can be applied in doing rapid reef assessment. Moreover, it is an objective method and can generate continuous data that can be used in statistical analysis. However, it could lose its fine resolution under extremely turbid conditions. By converting the photos to an 8-bit grey scale, some colour details may also be lost. It will be possible to improve the grey scale of the photos to address this limitation, but a better imaging analysis programme will be needed.

Some coral species exhibit different types of colours. This could affect the estimates for percent whiteness especially if these colours are highly contrasting (e.g. between blue and brown). This could lessen the utility of this technique in evaluating colour change in these corals. However, if different colours are a consistent feature of the species, it is possible to develop species and colour specific conversion scheme during image processing. While it will be a bit more time consuming, it is doable.

It is recognized that under sunny conditions ($>1350 \mu\text{mol s}^{-1} \text{m}^{-2}$), the values of percent whiteness measured will be different from that under cloudy conditions. Similar findings were also made in phenological research done in forest ecosystems (Sonntag et al., 2012). However, very intense levels of irradiance seldom occur underwater in natural coral reefs and the deviation could be avoided by taking photo measurements during early morning or later in the afternoon. In addition, colour source may also affect the accuracy of the percent whiteness measured. As reflected in the present experiment, blue colour appears to be least sensitive and would result in a lower detection of percent whiteness when compared to other colours. This may be because most image sensors used in digital cameras are less sensitive to blue colour and hence poor image resolution will result in the presence of extra blue light/absence of red and green light. External light source, such as flash system of the camera, could provide extra white lights (equal portions of red, green and blue radiations) to overcome the issue of poor image resolution and allow this technique to detect colour change of a coral colony in an objective way under different spatial and temporal conditions.

The choice of digital camera may also affect the percent whiteness calculated as image processing is usually carried out within the camera (Macfarlane et al., 2014). However, Sonntag et al. (2012) revealed that the choice of digital camera is of secondary importance when doing repeated photography for phenological research in forest ecosystems. Key changes could still be easily detected and comparable among different cameras. Such should likely be the case when taking images of the same coral colony for colour change (e.g. coral bleaching) evaluation repeatedly. If subtle changes are to be detected (e.g. change in coral health/colour over a period of time), it is recommended that a specific camera with a standard protocol be implemented throughout the whole sampling period in order to achieve the best comparable results.

All in all, any method would have its own weakness and limitations. The advantages of employing this technique as a general tool to assess coral colour change and the degree of coral bleaching should outweigh some of its limitations. More extensive application of this technique under different conditions for different coral species may provide further refinement to make it a readily usable and objective tool for rapid and easy assessment of coral health.

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